



Enea MANZI

Curriculum Vitae

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🎓 | Education

- 09/2024 - 07/2026 ○ **Master's Degree in Cyber Security, University of Milan, (Milan, IT)**
Thesis: "Design and Implementation of an API-Agnostic Tool for REST API Security Assurance"
Supervisors: Marco Anisetti, Claudio Agostino Ardagna
Expected Graduation: 07/2026
- 09/2021 - 10/2024 ○ **Bachelor's Degree in System and Network Security, University of Milan, (Milan, IT)**
Thesis: "Design and Implementation of a Framework for Evaluating Distributed System"
Supervisors: Marco Anisetti, Filippo Berto
Final Grade: 110/110 cum laude
- 09/2016 - 06/2021 ○ **High School Diploma in Computer Science, IIS Badoni, (Lecco, IT)**
Final Grade: 93/100

👜 | Work Experience

- 12/2025 - 07/2026 ○ **Curricular Internship for Master's Thesis, SESAR Lab, University of Milan, (Milan, IT)**
Secure Service-oriented Architectures Research Lab (SESAR Lab)
Designed and implemented APIGuard Assurance, an API-agnostic, contract-driven tool for automated REST API security assessment based on a taxonomy of eight domains and specified oracles, validated against a real-world target.
- 03/2024 - 09/2024 ○ **Curricular Internship for Bachelor's Thesis, SESAR Lab, University of Milan, (Milan, IT)**
Secure Service-oriented Architectures Research Lab (SESAR Lab)
Contributed to the design and implementation of an Assurance Engine for the verification of non-functional properties (NFP) in distributed systems, built in Rust and deployed on Kubernetes, extending it with Elasticsearch integration.
- 01/2020 - 02/2020 ○ **WBL: web developer, DaTriK Solutions, (Casargo, IT)**
High school work-based-learning
Developed a web application using HTML, JavaScript, and jQuery as part of a high school work-based learning program (PCTO).

| Research Activities

12/2025 - 07/2026 ○

SESAR Lab, University of Milan, (Milan, IT)

Secure Service-oriented Architectures Research Lab (SESAR Lab)

- Investigated the limitations of existing REST API testing approaches in detecting semantic security violations, motivating an API-agnostic, contract-driven methodology that derives target knowledge exclusively from the OpenAPI Specification (OAS).
- Defined a taxonomy of eight security domains, aligned with the NIST SP 800-228 guidelines for API security, to structure systematic assessment coverage.
- Studied the "oracle problem" in automated security testing and proposed *specified oracles*, evaluation criteria derived from the domain-specific knowledge of each security control, traceable to OWASP categories and CWE codes.
- Investigated reproducibility requirements for security assessment in continuous development, designing a deterministic, DAG-based scheduling approach exposing semantic exit codes for CI/CD integration.
- Validated the methodology against a real-world target, Forgejo exposed through the Kong API gateway, confirming correct detection of security violations and deterministic, reproducible results.

03/2024 - 09/2024 ○

SESAR Lab, University of Milan, (Milan, IT)

Secure Service-oriented Architectures Research Lab (SESAR Lab)

- Investigated existing verification and monitoring approaches for distributed systems, contributing to the extension of an Assurance Engine with Elasticsearch as a pluggable monitoring backend.
- Studied and formalized contracts to jointly evaluate multiple non-functional properties (CIA triad, reliability, scalability, performance), validated experimentally on a Kubernetes testbed with minimal computational overhead.

| Projects

APIGuard *Master's Thesis Project (A.Y. 2025/2026)*

**Assurance – An
API-Agnostic
Tool for REST
API Security
Assurance**

A security assessment tool for REST APIs exposed through API gateways, designed to detect semantic vulnerabilities that traditional syntax-based scanners miss, by deriving all target knowledge directly from the OpenAPI Specification.

- Built in **Python 3.12** with **Pydantic** models for runtime data validation and a plugin-style test-discovery mechanism, allowing new security checks to be added without modifying the core engine.
- Verified the engine's own reliability before trusting its security verdicts, enforcing strict static typing (**mypy**) and automated quality scanning (**ruff**, **bandit**, **vulture**, **pip-audit**) to achieve full docstring coverage on the public API.
- Packaged with **Hatch**, verifying byte-reproducible builds (identical SHA-256 wheel hash across runs) and isolating an **AGPL-licensed** dependency behind an optional install extra to keep the core package free of copyleft propagation.
- Integrated external security scanners (**nuclei**, **testssl.sh**, **sslyze**) through a hierarchical connector architecture, keeping the core test logic decoupled from how each external tool's output is parsed.

- Event-Driven IoT Microservices Architecture (Kafka + Kong)** *Cloud Computing Technologies Course Project (A.Y. 2025/2026)*
- Development of a cloud-native Kubernetes infrastructure for IoT ingestion and monitoring, with a particular focus on **non-functional properties** such as scalability, fault tolerance, and edge-to-core security.
- Microservices architecture based on **Apache Kafka** in KRaft mode (ZooKeeper-less) for asynchronous decoupling and reliable message handling (*Zero Data Loss*).
 - Integration of **Kong API Gateway** to implement the *Gateway Offloading* pattern, centralizing Authentication (API Key), Rate Limiting, and Load Balancing at the edge.
 - Storage layer optimized on **MongoDB** using **Time Series Collections** and hybrid compression (LZ4/Zstd) to maximize storage efficiency and analytical performance.
 - Full **Kubernetes** orchestration managed via **Helm**, featuring automatic scalability (HPA) and advanced security (Secrets, TLS/SASL) to meet non-functional requirements.
- FTP Client and Server** *Computer Networks Course Project (A.Y. 2022-2023)*
- Developed a multi-threaded **FTP-compliant** client-server application in **C** for file transfer, fully interoperable with **FileZilla**.
- Architecture: Multi-threaded server based on **Socket Programming** for concurrent session management.
 - Data Modes: Native support for both Active (PORT) and Passive (PASV) data transfer modes.
 - Protocol Management: Implemented essential *FTP commands* (e.g., USER, PASS, LIST, RETR, STOR, CWD...)
 - **Key skills: C, socket programming, multithreading, network protocols, FTP, Git.**
- Exhibition Web App** *Web and Mobile Programming Course Project (A.Y. 2021-2022)*
- Built a dynamic web application serving as a digital portfolio for displaying multimedia content.
- Developed the backend with *Node.js/Express* and the frontend with *HTML5, CSS3, JavaScript/jQuery*.
 - Integrated *MongoDB* for document-oriented storage and *AJAX* for asynchronous client-server communication (GET, POST, DELETE).
 - Managed source code and collaboration using **Git** and **GitHub**.
 - **Key skills: Node.js, Express, MongoDB, AJAX, REST APIs, jQuery, Git.**

| Publications

TODO: exact title Enea Manzi, Marco Anisetti, Claudio A. Ardagna
ITADATA 2026 (submitted)
 TODO: draft below, to refine

| Languages

Italian Mother tongue

English Intermediate *Listening B2 - Reading B2 - Writing B2 - Spoken Production B2 - Spoken Interaction B2*

| Skills

Programming Languages *C, Java, Python, SQL, Web (HTML, JavaScript/Node.js), OOP Concepts, low-level programming*

Database *MySQL, Database Design, Implementation and Administration*

Architectures & Software *Docker, Kubernetes, microservices architectures, REST API, Git, Kafka, Kong API Gateway, CI/CD*

Operating Systems	Fresh OS installation, Linux Administration and Shell, Windows
Networking & Security	Network Protocols, ICT Network Security Risk, Secure Coding Practices, API Security, Continuous Security Assurance
Tools	Visual Studio Code, GitHub

| Soft Skills

Thinking	analytically, critically, proactively
Problems	address problems critically, solve problems/create solutions
Communication	cooperate with colleagues, work in teams, manage teamwork
Others	demonstrate curiosity, cope with stress, prioritise tasks

Bellano, June 25, 2026
Enea Manxi